Temperature scaling for CoBit on TinyGSM → GSM8K

All experiments and results, 11–13 October 2026 · CoBit-S (130M), its CFG variant and CoBit-M (462M) · GSM8K executable-code accuracy, all 1319 test problems · Isambard GH200 · prepared for Georgios

> **Main result.** Sigma-style score temperature (the score divided by τ at every noise level) works on CoBit once the sampler's last step is left unsharpened. On the main unguided CoBit-S model (500k) accuracy goes from **28.1% to 35.3%** at τ=0.3 (+7.1 [+5.1, +9.2] pts, paired 95% CI). On CoBit-M (250k) it goes from 31.5% to 39.3% (τ=0.3, +7.7 [+5.5, +9.9]). Recipe: EDM churn γ=0.41, τ≈0.3, last step untempered.

finding	evidence
Score temperature helps every checkpoint we tested	+7.1 (CoBit-S 500k), +7.7 (CoBit-M), +6.0 (CoBit-S CFG ckpt) pts, all CIs exclude 0
Classifier-free guidance adds nothing on top	best (w, τ) = (1, 0.2): 31.9%
Below τ≈0.7 the old sampler collapsed — two separate causes	churn: the last step (σ 0.157→0.002) corrupted bits; deterministic: mode/length collapse
Fix: untempered last step (or stop at σ_min=0.1)	churn works down to τ=0.2; invalid tokens back to 0
Lower γ widens the usable τ range but not accuracy	γ=0.1–0.2 stable at τ=0.1 (32–33%); best 33.6% < 35.3%
Ancestral (DDPM-style) sampler: stable but weaker	≤ 29.0% (starts at 16.6% at τ=1)
DeepMind's logit temperature: no real gain	best T=0.1: 29.5% (+1.4 [-0.6, +3.3] pts, n.s.)
σ_min=0.002 sends the sampler where the model was never trained	no training mass below σ≈0.17

Caveats up front: one sampling seed per cell (intervals are bootstrap over problems; differences are paired on identical problems and noise); the CoBit-M and CFG arms use the Karras σ grid while the main unguided arm uses the entropic grid, so absolute numbers are comparable only within an arm.

1 Three ways to apply a temperature

CoBit diffuses bits $x_0 \in \{0,1\}$ with a variance-exploding process $x = x_0 + \sigma\varepsilon$. The network gives the per-bit posterior mean D = sigmoid(learned logit + analytic matched filter), and the score is $s = (D - x)/\sigma^2$. The three temperatures we tested act on this in different places:

name	operation	where it acts
A1 “local” (our original)	$s \to \kappa(\sigma)\, s$, $\kappa = (v + \sigma^2)/(\tau v + \sigma^2)$, $v = 0.25$	≈1 at high σ, $1/\tau$ only as $\sigma \to 0$: sharpens only the end of the trajectory
Score temperature (Sigma, Yang et al. 2026)	$s \to s/\tau \Leftrightarrow D_\tau = x + (D - x)/\tau$	every σ, constant; extrapolates D beyond [0,1]
Logit temperature (DeepMind SDM, eq. 25)	token logits $\zeta \to \zeta/T$, $D_T = E_{\text{softmax}(\zeta/T)}[\text{code}]$	every σ; sharpens the prediction inside [0,1]

Your note of 11 Oct pointed out that our earlier negative result was for A1 only, which leaves the high-noise part of the trajectory untempered; everything below tests the global (Sigma) operation, and at the end DeepMind's. Code: `--score_temp_mode global` (CoBit), τ=1 bit-identical to the old sampler (unit-tested).

2 Setup

checkpoint	what it is	σ grid used
CoBit-S unguided, 500k	the main TinyGSM run (shared Drive last.pt, global_step 500000)	entropic
CoBit-S CFG, 425k	same recipe with 10% prompt dropout	Karras
CoBit-M, 250k	462M, unguided, half-way through a 500k cosine	Karras

- All 1319 GSM8K problems, one answer each, EMA weights, 1024 steps unless stated, seed 0.
- EDM churn γ=0.41 per step (“churn”) unless stated; “deterministic” = γ=0 (probability-flow ODE).
- Δ = paired difference on identical problems, bootstrap 95% CI. Programs that ran = share of generated programs that execute to a number; invalid tokens = generated ids outside the vocabulary.

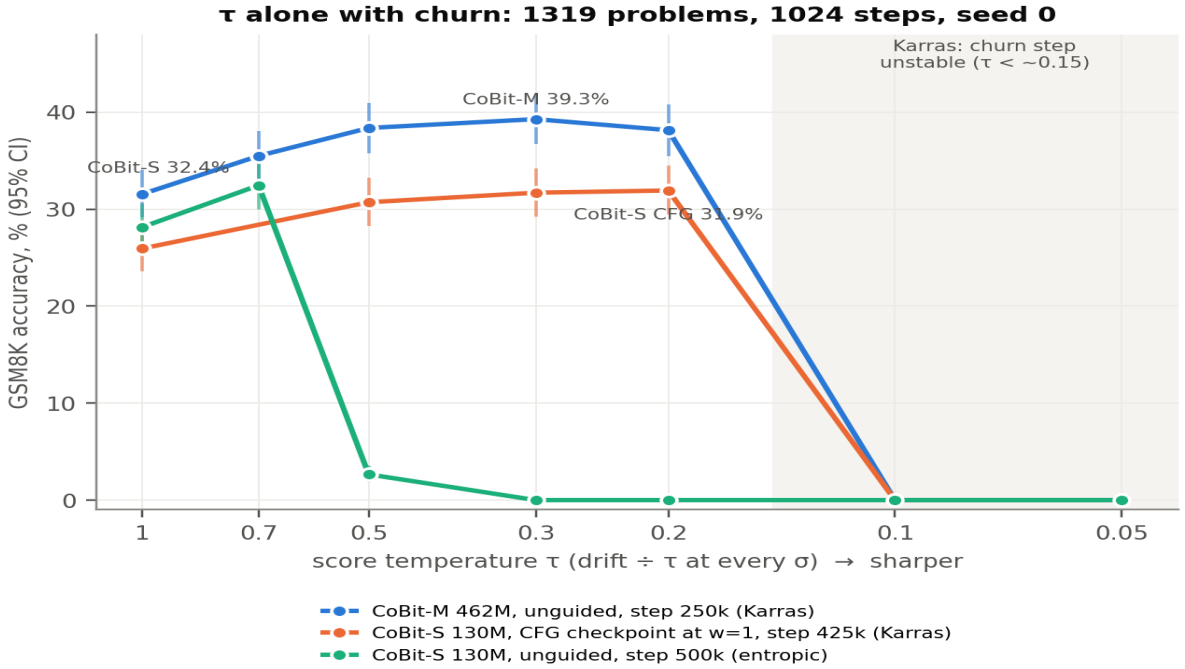
3 Before this study

Two earlier results bear on the question (from the June/September FKC work; 256 problems, so ±6 pts).

A1, 1024 steps		**whole-trajectory scaling (FKC drift-only, 512 steps)**	
κ-temperature τ (A1, late-σ only)	**accuracy %**	**τ = 1/β (whole trajectory)**	**accuracy %**
0.3	0.0	1.00	28.52
0.5	32.03	0.98	30.08
0.7	34.77	0.95	28.52
1.0	33.98	0.91	30.86
		0.87	30.47
		0.83	32.81
		0.77	32.03

A1 gave +0.8 pts at τ=0.7 then failed (0% at τ=0.3). The FKC drift-only runs were already constant scaling of the drift, but only for $\tau \geq 0.77$, at 512 steps: a +4 pt trend that was read at the time as a step-size effect. Neither tested global score temperature at strong sharpening.

4 Global score temperature on three checkpoints (12 Oct)



τ alone with churn, 1024 steps. Bars: 95% CIs. Note the CoBit-S 500k curve (entropic grid) collapses below τ=0.7 while the two Karras-grid curves hold to τ=0.2 — explained in Sections 6–8.

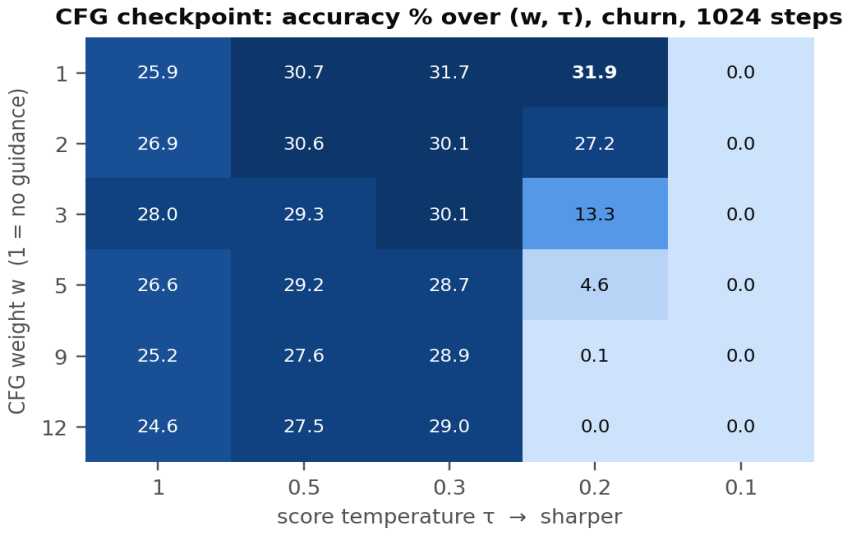
CoBit-M 462M (250k)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	31.5	[29.1, 34.0]	—	67.4
0.7	35.5	[32.9, 38.1]	+3.9 [+2.0, +5.9]	74.6
0.5	38.4	[35.8, 40.9]	+6.8 [+4.7, +8.9]	77.3
0.3	39.3	[36.7, 41.8]	+7.7 [+5.5, +9.9]	78.5
0.2	38.1	[35.5, 40.8]	+6.6 [+4.5, +8.8]	78.9
0.1	0.0	[0.0, 0.0]	-31.5 [-34.0, -29.1]	0.0
0.05	0.0	[0.0, 0.0]	-31.5 [-34.0, -29.1]	0.0

CoBit-S CFG checkpoint at w=1 (no guidance)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	25.9	[23.6, 28.3]	—	60.5
0.5	30.7	[28.3, 33.2]	+4.8 [+2.7, +6.8]	69.7
0.3	31.7	[29.2, 34.2]	+5.8 [+3.8, +7.8]	72.6
0.2	31.9	[29.4, 34.5]	+6.0 [+3.9, +8.1]	75.2
0.1	0.0	[0.0, 0.0]	-25.9 [-28.3, -23.6]	0.0
0.05	0.0	[0.0, 0.0]	-25.9 [-28.3, -23.6]	0.0

CFG × τ (CFG checkpoint)



Accuracy % per (w, τ); w=1 is the conditional model.

- τ does the work: best cell w=1, τ=0.2 (31.9%). CFG alone peaks at w=3 (28.0%), and every w>1 row at its best τ is at or below the w=1 row.
- Guidance moves the breaking point: at τ=0.2 accuracy falls from 31.9% (w=1) to 13.3% (w=3) and ≈0 for w≥9.
- Sigma's operating points (w=2, τ=0.1) and (1, 0.05) give 0% here: with EDM churn the drift ÷ τ makes every step unstable below τ≈0.15 (Section 6).

5 Experiment 1: the main unguided CoBit-S model

A checkpoint problem first

- The 425k file in the shared Drive folder scores 1.7% at τ=1 (churn, 1024 steps). Its EMA weights are complete, finite and were updating normally (an earlier note blamed a stale EMA; that was wrong). Its running weights do better (18.3% vs 4.9% at 512 steps), consistent with your recollection, but both are far below the 29.4% of June, which our handoff attributes to a main-run 425k that was never copied to the Drive folder.
- The Drive last.pt is the 500k checkpoint of the same run and is healthy, so Experiment 1 uses it.
- A silent fallback was found and fixed on the way: without the run's entropy tables, churn switched itself off with no warning; the launcher now refuses to run in that state.

Which σ grid? (500k EMA, τ=1, churn)

σ grid	accuracy %	95% CI	programs that ran %
entropic	28.1	[25.8, 30.6]	63.5
Karras	26.0	[23.7, 28.4]	59.3

Both reproduce your ~29%; entropic is used for this arm.

τ sweep with the original sampler (entropic grid)

churn γ=0.41, 1024 steps

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	28.1	[25.8, 30.6]	—	63.5
0.7	32.4	[29.9, 35.0]	+4.3 [+2.4, +6.2]	69.2
0.5	2.7	[1.8, 3.6]	-25.5 [-27.9, -23.1]	5.0
0.3	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0
0.2	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0
0.1	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0
0.05	0.0	[0.0, 0.0]	-28.1 [-30.6, -25.8]	0.0

deterministic (γ=0), 1024 steps

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	12.2	[10.5, 14.0]	—	31.5
0.7	24.7	[22.4, 27.1]	+12.5 [+10.3, +14.6]	63.1
0.5	4.4	[3.3, 5.5]	-7.8 [-9.8, -5.8]	47.2
0.3	0.3	[0.1, 0.6]	-11.9 [-13.7, -10.2]	12.7
0.2	0.4	[0.1, 0.8]	-11.8 [-13.6, -10.1]	23.6
0.1	0.9	[0.5, 1.4]	-11.3 [-13.1, -9.6]	40.8
0.05	1.1	[0.6, 1.7]	-11.1 [-13.0, -9.3]	52.6

churn γ=0.41, 512 steps (step-count control)

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	26.8	[24.4, 29.2]	—	60.0
0.7	31.5	[29.0, 34.0]	+4.8 [+2.7, +6.9]	68.8
0.5	2.2	[1.4, 3.0]	-24.6 [-27.0, -22.1]	5.5
0.3	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0
0.2	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0
0.1	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0
0.05	0.0	[0.0, 0.0]	-26.8 [-29.2, -24.4]	0.0

τ=0.7 helps (+4.3 [+2.4, +6.2] pts with churn; the deterministic sampler doubles), and 512 vs 1024 steps give the same gain, so it is not a step-size effect. But every setting collapses at τ ≤ 0.5.

6 Why it collapsed: the sampler, step by step

Every sampler we have is a step of the form below: the state moves towards the prediction D by a gain g and fresh noise is added. Score temperature multiplies the gain by 1/τ.

```
x' − D = (1 − g/τ) (x − D) + noise
```

- The step is **stable** only if g/τ < 2 and **does not overshoot** only if g/τ ≤ 1. When it overshoots, the state is reflected through D and the residual — including the noise just injected — is amplified by |1 − g/τ|.
- Gains: deterministic DDIM g = 1 − σ'/σ (≈0.01 per step on a 1024-step grid); EDM churn first raises σ by (1+γ) on every step, so g = 1 − σ'/((1+γ)σ) ≈ 0.29 at γ=0.41 regardless of the step count — the reason for the τ ≈ 0.15 wall on the Karras grid; the ancestral (DDPM) step g = 1 − (σ'/σ)², small.
- **The entropic grid ends with one huge step**, σ = 0.157 → 0.002 (g = 0.99). Any τ < 1 makes it overshoot; with churn, the reflected residual is large enough to flip already-decided bits.
- DeepMind's logit temperature cannot overshoot: it sharpens D inside [0,1] and leaves the step a convex combination, which is why their T=0.1 works with churn.

The two failure modes, from the real outputs (500k, entropic, 1024 steps)

sampler (500k EMA, entropic, 1024 steps)	τ	accuracy %	programs run %	invalid tokens %	median program length	body = 'return result' only
churn γ=0.41	1	28.1	63.5	0.00	166 chars	1%
churn γ=0.41	0.7	32.4	69.2	0.00	152 chars	1%
churn γ=0.41	0.5	2.7	5.0	1.39	150 chars	1%
churn γ=0.41	0.3	0.0	0.0	40.40	399 chars	0%
deterministic	1	12.2	31.5	0.00	158 chars	1%
deterministic	0.7	24.7	63.1	0.00	146 chars	1%
deterministic	0.5	4.4	47.2	0.00	74 chars	32%
deterministic	0.3	0.3	12.7	0.00	13 chars	60%

- **With churn**: invalid token ids appear (0 → 1.4 → 40%) while program length is unchanged — the right sentence with scrambled tokens. Bit corruption at the last step.
- **Deterministic**: never an invalid token, but programs shrink to *return result*. Constant 1/τ on an ODE is strongly mode-seeking, and ending early is the dominant mode; churn's noise counteracts it.

Predictions from an exact-posterior toy

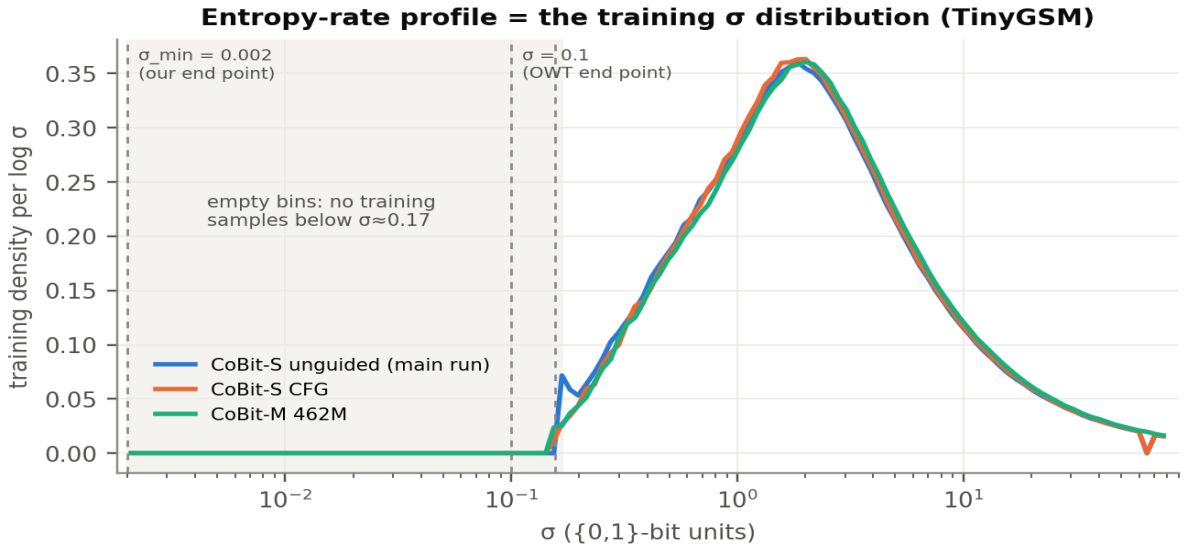
Independent bits with the exact posterior (no network), the real grids: share of bits whose readout flips the decision held one step earlier.

sampler (toy)	σ grid	last step	τ=0.5	τ=0.3	τ=0.2	τ=0.1
churn γ=0.41	entropic	sharpened	0.12%	8.1%	23%	diverges
churn γ=0.41	entropic	unsharpened	0	0	0	diverges
churn γ=0.41	karras	sharpened	0	0	0	diverges

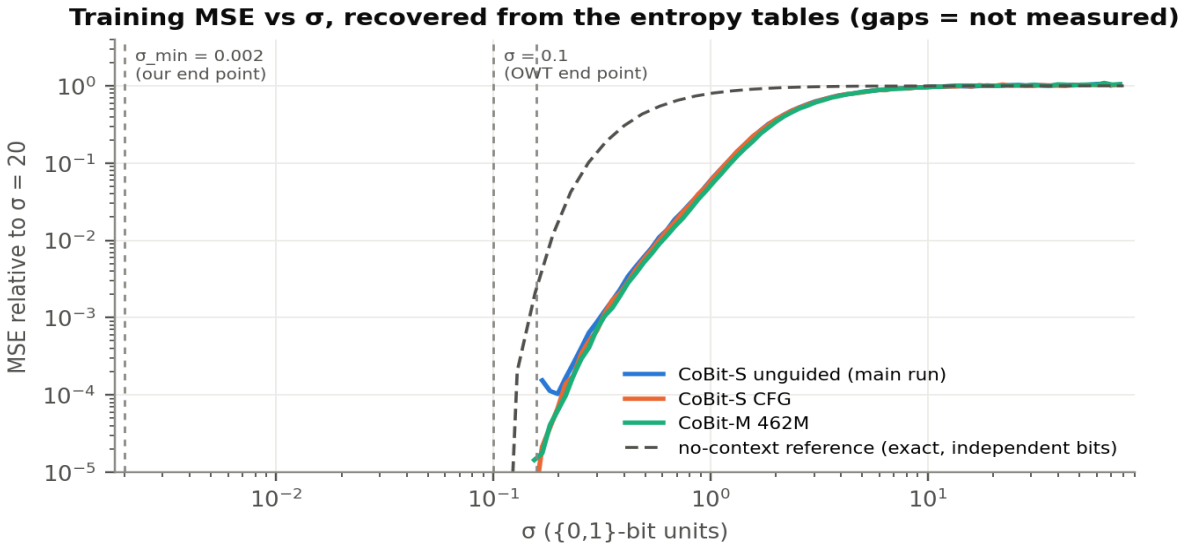
sampler (toy)	σ grid	last step	τ=0.5	τ=0.3	τ=0.2	τ=0.1
churn γ=0.2	entropic	sharpened	0.012%	2.9%	11%	36%
churn γ=0.2	entropic	unsharpened	0	0	0	0
churn γ=0.2	karras	sharpened	0	0	0	0
ancestral	entropic	sharpened	0	0.047%	0.95%	7.3%
ancestral	entropic	unsharpened	0	0	0	0
ancestral	karras	sharpened	0	0	0	0

Prediction: leaving the last step untempered removes the flips; lower γ and ancestral steps move the divergence wall to smaller τ. Section 8 tests this on the real model.

7 The entropy-rate profile and σ_min



Training σ distribution stored by each run. 54 of 128 bins are empty: after the 40k-step warm-up the model is essentially never trained below σ≈0.17.

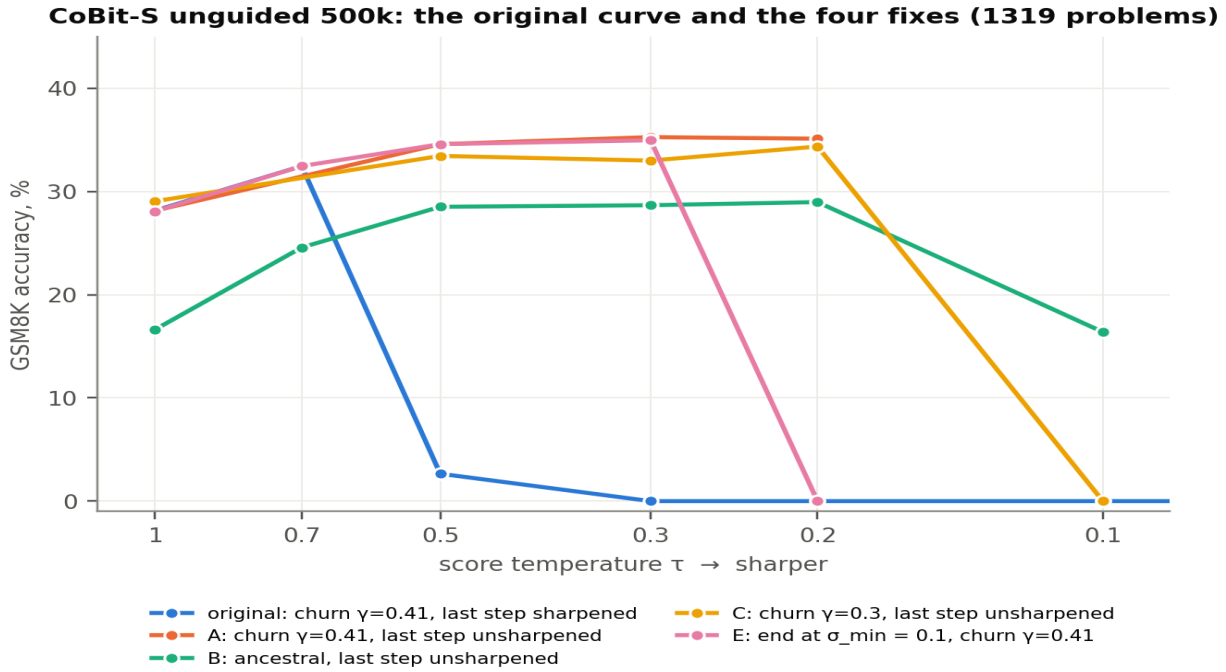


Training MSE per σ recovered from the same tables (scaled to 1 at σ=20); dashed: exact MMSE of an isolated bit with no context. TensorBoard on Isambard holds only loss/LR/σ_data, so the tables are the per-σ record.

- We sample to σ_min = 0.002 (EDM's default, {0,1}-bit units; not 2e-4). The grid's last step jumps from 0.157 to 0.002, into the untrained region.
- No numerical blow-up: below σ≈0.13 the matched-filter logit sits at its ±30 clip, and the drift stays O(1) in fp32. The risk is the learned logit evaluated where it was never trained.
- Even an isolated bit is decided by σ ≈ 0.1, so nothing is gained by going lower. Ending at σ_min = 0.1 (as on OWT) cuts the last step's gain from 0.99 to 0.36.

8 Fixing the collapse (13 Oct)

All cells: CoBit-S unguided 500k EMA, entropic grid, 1024 steps, 1319 problems, seed 0, paired with the cells of Section 5. A: the last step (onto σ_min, the readout) uses the untempered drift. B: the VE ancestral sampler (exact Gaussian bridge, no churn). C: lower churn γ, last step unsharpened. D: DeepMind's logit temperature T (softmax over the valid codewords with the full logit, every step). E: the trajectory ends at σ_min = 0.1 instead of 0.002 (see entropy_profile.pdf).



Accuracy per cell (each 1319 problems). A's τ=1 point is the original τ=1 cell (the two settings coincide at τ=1).

A — unsharpened last step

sampler	τ	last step sharpened: acc / run / invalid	unsharpened: acc / run / invalid	unsharpened Δ vs τ=1 (paired)
churn γ=0.41	0.5	2.7 / 5.0 / 1.39	34.6 / 72.6 / 0.00	+6.4 [+4.4, +8.5]
churn γ=0.41	0.3	0.0 / 0.0 / 40.40	35.3 / 75.4 / 0.00	+7.1 [+5.1, +9.2]
churn γ=0.41	0.2	0.0 / 0.0 / 41.06	35.1 / 74.0 / 0.00	+7.0 [+4.9, +9.1]
deterministic	0.5	4.4 / 47.2 / 0.00	4.4 / 47.2 / 0.00	-7.8 [-9.8, -5.8]
deterministic	0.3	0.3 / 12.7 / 0.00	0.3 / 12.7 / 0.00	-11.9 [-13.7, -10.2]
deterministic	0.2	0.4 / 23.6 / 0.00	0.4 / 23.6 / 0.00	-11.8 [-13.6, -10.1]

B — ancestral sampler (no churn), last step unsharpened

τ	accuracy %	95% CI	Δ vs τ=1, paired (95% CI)	programs that ran %
1	16.6	[14.6, 18.7]	—	41.8
0.7	24.6	[22.3, 26.9]	+8.0 [+6.1, +9.9]	59.0
0.5	28.5	[26.1, 30.9]	+11.9 [+9.9, +14.1]	66.6
0.3	28.7	[26.2, 31.1]	+12.1 [+9.9, +14.3]	69.1
0.2	29.0	[26.5, 31.4]	+12.4 [+10.0, +14.8]	71.9
0.1	16.4	[14.5, 18.4]	-0.2 [-2.7, +2.2]	78.8

C — churn γ × τ, last step unsharpened (accuracy %)

γ \ τ	1	0.5	0.3	0.2	0.1	best Δ vs own τ=1
0.1	26.8	33.0	33.6	32.4	32.0	τ=0.3: +6.7 [+4.5, +9.0]
0.2	28.0	33.3	33.4	33.5	33.3	τ=0.2: +5.5 [+3.3, +7.7]
0.3	29.0	33.4	33.0	34.3	0.0	τ=0.2: +5.3 [+3.1, +7.4]
0.41 (orig.)	28.1	2.7	0.0	0.0	0.0	sharpened last step

E — end the trajectory at σ_min = 0.1

end point / last step	τ	accuracy %	programs run %	invalid tokens %	Δ vs τ=1 at σ_min 0.1
σ_min 0.1, sharpened	1	28.1	63.5	0.00	—
σ_min 0.1, sharpened	0.7	32.4	69.2	0.00	+4.4 [+2.5, +6.2]
σ_min 0.1, sharpened	0.5	34.6	72.6	0.00	+6.5 [+4.5, +8.6]
σ_min 0.1, sharpened	0.3	35.0	74.9	0.00	+6.9 [+4.9, +8.9]
σ_min 0.1, sharpened	0.2	0.0	0.0	13.60	-28.1 [-30.5, -25.7]
σ_min 0.1, unsharpened	0.3	35.3	75.4	0.00	+7.2 [+5.2, +9.2]
σ_min 0.1, unsharpened	0.2	35.1	74.0	0.00	+7.1 [+4.9, +9.2]

D — DeepMind logit temperature (churn γ=0.41)

logit temperature T	accuracy %	programs run %	invalid tokens %	Δ vs churn τ=1 (paired)
0.5	28.1	61.6	0.00	-0.1 [-1.9, +1.7]
0.3	28.5	64.3	0.00	+0.4 [-1.4, +2.1]
0.1	29.5	61.5	0.00	+1.4 [-0.6, +3.3]

Observations

- **The churn collapse was the last step.** With it unsharpened, churn γ=0.41 keeps working to τ=0.2 (34.6 / 35.3 / 35.1% at τ=0.5 / 0.3 / 0.2, against 2.7 / 0.0 / 0.0% before) and invalid tokens fall back to 0. Best: τ=0.3, +7.1 [+5.1, +9.2] pts over τ=1.
- **The deterministic collapse is not.** Unsharpening the last step leaves it unchanged (4.4% at τ=0.5): it is the mode/length collapse of Section 6, as predicted.
- **Ancestral** is stable to τ=0.2 (29.0%, +12.4 [+10.0, +14.8] pts over its own τ=1) but starts much lower (16.6% at τ=1, little noise) and drops back at τ=0.1 (16.4%). It never beats churn.
- **Lower γ extends the range**: at τ=0.1, γ=0.1 / 0.2 score 32.0 / 33.3% while γ=0.3 and 0.41 diverge, as the stability rule predicts. But no γ beats γ=0.41 with an unsharpened last step at its best τ; lower γ buys range, not accuracy.
- **Ending at σ_min = 0.1** alone recovers τ=0.5–0.3 (34.6 / 35.0%) but a sharpened last step still corrupts bits at τ=0.2 (0.0%). With the last step unsharpened, ending at 0.1 or 0.002 gives identical answers: once that step is untempered, it decides nothing.
- **DeepMind's logit temperature** is stable at every T (no invalid tokens), but helps far less: best T=0.1, 29.5% (+1.4 [-0.6, +3.3] pts over churn τ=1).

One seed throughout; the best cells should get seeds 1–2 before the numbers are quoted externally.

9 Conclusions and recommendation

- **Use global score temperature with an untempered last step.** Recommended setting for CoBit on TinyGSM: EDM churn γ=0.41, τ≈0.3, last step unsharpened (or stop at σ_min=0.1). Gains of 6–8 pts on all three checkpoints; best 35.3% for CoBit-S 500k.
- **The collapse below τ=0.7 was the sampler, not the model or the readout**: one oversized final step (churn arm) and ODE mode collapse (deterministic arm). The τ ≈ 0.15 wall with churn is a stability limit of the churn step itself.
- **CFG does not add to temperature** on our CFG checkpoint, whose unconditional branch comes from 10% dropout on a 130M model; this is not a fair test of Sigma's CFG.
- **Lower γ** makes very low τ usable (τ=0.1) but did not raise the best accuracy; **ancestral** steps are stable but start too low; **DeepMind's logit temperature** gave no significant gain here.
- **σ_min** should be 0.1, not 0.002: below ≈0.17 the model was never trained and nothing is left to decide.

Suggested next steps

- Seeds 1–2 on the best cells of each arm (≈1.5 GH200-h) before quoting numbers externally.
- Make σ_min = 0.1 and the untempered last step the default sampler settings for TinyGSM.
- Give the training σ schedule a floor so low-σ bins are not left empty.
- Carry over to rolling-CoBit: its sampler already reads out the final step untempered; its score- and logit-temperature sweeps are ready.

Appendix — runs, compute and code

batch	cells	what	GH200-h
12 Oct sweep	59 + 3 gates	τ on three checkpoints; (w, τ) grid; first Exp. 1 attempt (425k)	27.2
12 Oct reruns	14 + 2	Exp. 1 churn arm with entropy tables; raw-weight diagnostic	3.2
12 Oct Exp. 1 redo	21 + 2 anchors	500k checkpoint, entropic vs Karras anchor	5.3
13 Oct blocks A–E	38	unsharpened last step, ancestral, γ × τ, logit temperature, σ_min = 0.1	10.6

Total ≈ 46 GH200-h. Results: results/scoretemp/eval_gh200, anchor500/, diag/, blocks_gh200/. Drivers: scripts/hpc/arch/scoretemp_gh200.sh and scoretemp_blocks.sh; launcher scoretemp_eval.slurm. New sampler options (uncommitted, unit-tested): `--score_temp_mode global`, `--score_temp_final unsharpened`, `--sampler_kind ancestral`, `--sigma_min`; the EM sampler now accepts score temperature. Companion notes: report_score_temperature.pdf (full tables), sampler_math.pdf, entropy_profile.pdf. Built by docs/pdf_reports/build_temperature_study.py.